

Section 2.4 Matrix Inverses

Exercise 2.4.2

Find the inverse of each of the following matrices.

e. $A = \begin{bmatrix} 1 & 0 & -1 \\ 3 & 2 & 0 \\ -1 & -1 & 0 \end{bmatrix} \begin{array}{l} \left[\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & 0 & 0 \\ 3 & 2 & 0 & 0 & 1 & 0 \\ -1 & -1 & 0 & 0 & 0 & 1 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & -2 & -3 & 3 & -1 & 0 \\ 0 & -1 & -1 & 1 & 0 & 1 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 1 & 0 & 1 \\ 0 & -2 & -3 & 3 & -1 & 0 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 0 & -1 \\ 0 & -2 & -3 & 3 & -1 & 0 \end{array} \right]$

$$\begin{aligned} R_2 &= R_2 \\ R_2 &= R_2 - 2R_1 \end{aligned}$$

$$\begin{aligned} R_0 &= R_0 + R_2 \\ R_1 &= R_1 - R_2 \end{aligned}$$

$$\begin{aligned} R_1 &= 3R_0 - R_1 \\ R_2 &= R_2 + R_0 \end{aligned}$$

$$\text{SWAP}(R_2, R_1)$$

$$\Rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & -1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 0 & -1 \\ 0 & 0 & 1 & -1 & 1 & 2 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & 2 \\ 0 & 1 & 0 & 0 & -1 & -3 \\ 0 & 0 & 1 & -1 & 1 & 2 \end{array} \right], \text{ hence } A^{-1} = \begin{bmatrix} 0 & 1 & 2 \\ 0 & -1 & -3 \\ -1 & 1 & 2 \end{bmatrix}$$

i. $\begin{bmatrix} 3 & 1 & 2 \\ 1 & -1 & 3 \\ 1 & 2 & 4 \end{bmatrix} \begin{array}{l} \left[\begin{array}{ccc|ccc} 3 & 1 & 2 & 1 & 0 & 0 \\ 1 & -1 & 3 & 0 & 1 & 0 \\ 1 & 2 & 4 & 0 & 0 & 1 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|ccc} 1 & -1 & 3 & 0 & 1 & 0 \\ 3 & 1 & 2 & 1 & 0 & 0 \\ 1 & 2 & 4 & 0 & 0 & 1 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|ccc} 1 & -1 & 3 & 0 & 1 & 0 \\ 0 & 4 & -7 & 1 & -2 & -1 \\ 0 & 3 & 1 & 0 & -1 & 1 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|ccc} 1 & -1 & 3 & 0 & 1 & 0 \\ 0 & 1 & -8 & 1 & -2 & -1 \\ 0 & 3 & 1 & 0 & -1 & 1 \end{array} \right]$

$$\text{SWAP}(R_0, R_1)$$

$$R_1 = R_1 - 3R_0$$

$$R_2 = R_2 - R_0$$

$$R_0 = R_0 + R_1$$

$$R_2 = R_2 - 3R_1$$

$$R_1 = R_1 - R_2$$

$$\Rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & -5 & 1 & -1 & -1 \\ 0 & 1 & -8 & 1 & -2 & -1 \\ 0 & 0 & 25 & -3 & 5 & 4 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & -5 & 10/25 & 0 & -5/25 \\ 0 & 1 & -8 & 1/25 & -10/25 & 7/25 \\ 0 & 0 & 1 & -3/25 & 5/25 & 4/25 \end{array} \right] \Rightarrow A^{-1} = \frac{1}{25} \begin{bmatrix} 10 & 0 & -5 \\ 1 & -10 & 7 \\ -3 & 5 & 4 \end{bmatrix}$$

$$R_0 + R_1 = R_0$$

$$R_2 - 3R_1 = R_2$$

$$\frac{1}{25} R_2$$

$$R_0 = R_0 + 5R_2$$

$$R_1 = R_1 + 8R_2$$

Why does this work/does what?

Exercise 2.4.3 ✓

In each case, solve the systems of equations by finding the inverse of the coefficient matrix.

d.

$$\begin{aligned} x + 4y + 2z &= 1 \\ 2x + 3y + 3z &= -1 \\ 4x + y + 4z &= 0 \end{aligned} = \begin{bmatrix} 1 & 4 & 2 & 1 \\ 2 & 3 & 3 & -1 \\ 4 & 1 & 4 & 0 \end{bmatrix} = A, \text{ finding } A^{-1}$$

$$\Rightarrow \left[\begin{array}{ccc|ccc} 1 & 4 & 2 & 1 & 0 & 0 \\ 2 & 3 & 3 & 0 & 1 & 0 \\ 4 & 1 & 4 & 0 & 0 & 1 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|ccc} 1 & 4 & 2 & 1 & 0 & 0 \\ 0 & 5 & 1 & 2 & -1 & 0 \\ 0 & 15 & 4 & 4 & 0 & -1 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|ccc} 1 & 4 & 2 & 1 & 0 & 0 \\ 0 & 0 & -1 & 2 & -3 & 1 \\ 0 & 0 & -1 & 2 & -3 & 1 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|ccc} 1 & 4 & 2 & 1 & 0 & 0 \\ 0 & 1 & \frac{1}{5} & \frac{2}{5} & -\frac{1}{5} & 0 \\ 0 & 0 & -1 & 2 & -3 & 1 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|ccc} 1 & 4 & 2 & 1 & 0 & 0 \\ 0 & 1 & \frac{1}{5} & \frac{2}{5} & -\frac{1}{5} & 0 \\ 0 & 0 & -1 & 2 & -3 & 1 \end{array} \right]$$

$R_1 = 2R_0 - R_2$
 $R_2 = 4R_0 - R_2$
 $R_2 = 3R_1 - R_2$
 $R_3 =$
 $R_2 = 3R_1 - R_2$
 $R_1 = \frac{1}{5}R_1$
 $R_1 = R$

$$\Rightarrow \left[\begin{array}{ccc|ccc} 1 & 4 & 2 & 1 & 0 & 0 \\ 0 & -5 & -1 & -2 & 1 & 0 \\ 0 & 0 & -1 & 2 & -3 & 1 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|ccc} 1 & 4 & 2 & 1 & 0 & 0 \\ 0 & 1 & \frac{1}{5} & \frac{2}{5} & -\frac{1}{5} & 0 \\ 0 & 0 & -1 & 2 & -3 & 1 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|ccc} 1 & 4 & 0 & 5 & -6 & 2 \\ 0 & -5 & 0 & -4 & 4 & -1 \\ 0 & 0 & -1 & 2 & -3 & 1 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|ccc} 1 & 4 & 0 & 5 & -6 & 2 \\ 0 & 1 & 0 & 4/5 & -4/5 & 1/5 \\ 0 & 0 & -1 & -2 & 3 & -1 \end{array} \right]$$

$R_2 = 3R_1 - R_2$
 $R_1 = R_0 - R_1$
 $-R_3, \frac{1}{5}R_2$
 $R_2 + R_3 = R_2$
 $R_1 + 2R_3 = R_1$
 $-R_3$
 $-\frac{1}{5}R_2$

$$A^{-1} \Rightarrow \begin{bmatrix} 9/5 & -14/5 & 6 \\ 4/5 & -4/5 & 11/5 \\ -2 & 3 & -1 \end{bmatrix}, \text{ then } \frac{1}{5} \begin{bmatrix} 9 & -14 & 6 \\ 4 & -4 & 1 \\ -10 & 15 & -5 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \Rightarrow \frac{1}{5} (9 + 14 + 0) = 23/5 = x$$

$$\frac{1}{5} (4 + 4) = 8/5 = y$$

$$\frac{1}{5} (-10 - 15) = -5 = z$$

Exercise 2.4.5

Find A when:

$$\text{d. } (I - 2A^T)^{-1} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \Rightarrow I - 2A^T = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} \Rightarrow -2A^T = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ \Rightarrow 2A^T = \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix} \Rightarrow A^T = \begin{bmatrix} 0 & -\frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} = A^T$$

B/c matrix = 2×2 , both $1/2$, no change, $\therefore$

$$A = \begin{bmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix} //$$

$$\frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\text{f. } \left(\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} A \right)^{-1} = \begin{bmatrix} 1 & 0 \\ 2 & 2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} A = \begin{bmatrix} 1 & 0 \\ 2 & 2 \end{bmatrix} \Rightarrow A = \begin{bmatrix} 1 & 0 \\ 2 & 2 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 \\ 2 & 2 \end{bmatrix} \\ \left\{ \frac{1}{1-0} \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ 2 & 2 \end{bmatrix} \right. \\ \left. = \begin{bmatrix} 1+0 & 0(0) \\ -2-1 & 0+2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -3 & 2 \end{bmatrix} = A \right. //$$

$$\text{h. } (A^{-1} - 2I)^T = -2 \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\bar{A}^{-1} - 2I = -2 \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\bar{A}^{-1} = \begin{bmatrix} -2 & -2 \\ -2 & 0 \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & -2 \\ -2 & 2 \end{bmatrix}$$

$$A = \frac{1}{2-4} \begin{bmatrix} 2 & 2 \\ 2 & 0 \end{bmatrix} \Rightarrow -\frac{1}{2} \begin{bmatrix} 2 & 2 \\ 2 & 0 \end{bmatrix}$$

$$\therefore A = -\frac{1}{2} \begin{bmatrix} 2 & 2 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} -1 & -1 \\ -1 & 0 \end{bmatrix} //$$

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Exercise 2.5.8

In each case factor A as a product of elementary matrices.

b. $A = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

SWAP (R_0, R_1) $R_1 = R_1 - 2R_0$ $-R_1$ $R_0 = R_0 - 2R_1$

1 $\rightarrow$ 2 $\rightarrow$ 3 $\rightarrow$ 4 $\rightarrow$

$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$\Rightarrow I E_i + \dots I E_{i+m}$

$\Rightarrow \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = A$, want A^{-1}

$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = A^{-1}$

d. $A = \begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & 4 \\ -2 & 2 & 15 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & 4 \\ 0 & 2 & 9 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$R_2 = R_2 - 2R_1$ $R_1 = R_1 - 4R_2$ $R_0 = R_0 + 3R_2$

$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = A^{-1}$

$A^{-1} \Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Exercise 2.5.6

In each case find an invertible matrix U such that $UA = R$ is in reduced row-echelon form, and express U as a product of elementary matrices.

Same deal but A^{-1}
R-REF
↓
d

b. $A = \begin{bmatrix} 1 & 2 & 1 & 1 & 0 \\ 5 & 12 & -1 & 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 & 1 & 1 & 0 \\ 0 & 2 & -6 & -5 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 & 1 & 1 & 0 \\ 0 & 1 & -3 & -5/2 & 1/2 \end{bmatrix}$ Same idea; find A^{-1} , but save each step

$R_1 \leftarrow R_1 - 5R_2$ $\frac{1}{2}R_2$ $R_2 \leftarrow R_2 - 2R_1$ $\Rightarrow \begin{bmatrix} 1 & 0 & 7 & 6 & -1 \\ 0 & 1 & -3 & -5/2 & 1/2 \end{bmatrix}$ Generate E : steps, $E = IB$, where B = steps, so B is operation performed.

$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$\begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$

R-REF $\Rightarrow \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & -3 \end{bmatrix}$

~~Inverted~~ $A^{-1} = \begin{bmatrix} 6 & -1 \\ -5/2 & 1/2 \end{bmatrix}$

c. $A = \begin{bmatrix} 1 & 2 & -1 & 0 \\ 3 & 1 & 1 & 2 \\ 1 & -3 & 3 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & -5 & 4 & 2 \\ 0 & -5 & 4 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & -5 & 4 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 2 & -1 & 1 \end{bmatrix}$

① $R_1 = R_1 - 3R_2$

③ $R_2 = R_2 - R_1$

② $R_2 = R_2 - R_1$

$\Rightarrow \begin{bmatrix} 1 & 2 & -1 & 0 \\ 0 & 1 & -4/5 & -2/5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 3/5 & -1/5 & 0 \\ 2 & -1 & 1 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} 1 & 0 & 3/5 & 4/5 \\ 0 & 1 & -4/5 & -2/5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} -1/5 & 2/5 & 0 \\ 3/5 & -1/5 & 0 \\ 2 & -1 & 1 \end{bmatrix}$

$A = \begin{bmatrix} 1 & 0 & 3/5 & 4/5 \\ 0 & 1 & -4/5 & -2/5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$
in R-REF

④ $\frac{1}{5}R_1$

⑤ $R_3 = R_3 - 2R_1$

$IE, \dots, IE_5 \Rightarrow$

$U = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1/5 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
⑤ ④ ③ ② ①

Ref, sig factor, row
permitted 2 row operations; $A \rightarrow C$?

Exercise 2.5.3

Let $A = \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix}$ and $C = \begin{bmatrix} -1 & 1 \\ 2 & 1 \end{bmatrix}$.

SWAP $\Rightarrow$
 $R_1 - R_0 = R_1$?

- a. Find elementary matrices E_1 and E_2 such that $C = E_2 E_1 A$.

E_1
 $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix}$
 $\text{SWAP}(R_1, R_0) \mid R_1 = R_1 - R_0$
 $\begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} -1 & 1 \\ 1 & 2 \end{bmatrix} \Rightarrow \begin{bmatrix} -1 & 1 \\ 2 & 1 \end{bmatrix}$
 $\uparrow$
 C

$E_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
 $E_2 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

$\therefore \text{ans}$
 $E_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
 $E_2 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

no need check!
 but if so, $\Rightarrow E_2 \neq E_1 \neq A$
 follow order!

Section 2.5 Elementary Matrices

Exercise 2.5.2

In each case find an elementary matrix E such that $B = EA$.

- 1) -gen I
- 2) row op make $A \rightarrow B$?
- 3) apply $\uparrow$ to $\textcircled{1}I$
- 4) answer

$$B \Rightarrow AE?$$

a. $A = \begin{bmatrix} 2 & 1 \\ 3 & -1 \end{bmatrix}, B = \begin{bmatrix} 2 & 1 \\ 1 & -2 \end{bmatrix} \quad E$

$\rightarrow$
 $R_1 = R_1 - R_0 \Rightarrow B$

$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$, $\begin{bmatrix} 2 & 1 \\ 3 & -1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 2 & 1 \\ 2 & -1 \end{bmatrix}$
 $\rightarrow$ swap, Rem $EA \neq AE!!$
 $\therefore E = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$ $\checkmark$

c. $A = \begin{bmatrix} 1 & 1 \\ -1 & 2 \end{bmatrix}, B = \begin{bmatrix} -1 & 2 \\ 1 & 1 \end{bmatrix}$

$\rightarrow$
SWAP (R_0, R_1)

$\Rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 0-1, 2+0 \\ 1+0, 1+0 \end{bmatrix} = \begin{bmatrix} -1 & 2 \\ 1 & 1 \end{bmatrix} \therefore E = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \checkmark$

e. $A = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}, B = \begin{bmatrix} -1 & 1 \\ -1 & 1 \end{bmatrix}$

$-R_1$

$\Rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \times \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} -1+0, 1 \\ -1, -1 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ -1 & 1 \end{bmatrix}$

Go by column
to clear to R-REF

Exercise 2.4.17

If $c \neq 0$, find the inverse of $\begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 2 \\ 0 & 2 & c \end{bmatrix}$ in terms of c .

$$\begin{aligned} \left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 2 & -1 & 2 & 0 & 1 & 0 \\ 0 & 2 & c & 0 & 0 & 1 \end{array} \right] &\Rightarrow \left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & -1 & 0 & -2 & 1 & 0 \\ 0 & 2 & c & 0 & 0 & 1 \end{array} \right] &\Rightarrow \left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & -1 & 0 & -2 & 1 & 0 \\ 0 & 0 & c & 4 & -2 & 1 \end{array} \right] \\ &\begin{array}{l} R_1 = R_0 - R_1 \\ -R_2 \\ R_0 = R_1 + R_0 \\ R_2 = R_2 - 2R_1 \end{array} &\Rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & -1 & 1 & 0 \\ 0 & 1 & 0 & -2 & 1 & 0 \\ 0 & 0 & c & 4 & -2 & 1 \end{array} \right] &\Rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & -1 & 1 & 0 \\ 0 & 1 & 0 & -2 & 1 & 0 \\ 0 & 0 & 1 & 4/c & -2/c & 1/c \end{array} \right] \\ &\begin{array}{l} \frac{1}{c} R_2 \\ \uparrow \\ \text{so can set I on LHS} \end{array} \end{aligned}$$

$$\Rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 - \frac{4}{c} & 1 + \frac{2}{c} & -\frac{1}{c} \\ 0 & 1 & 0 & -2 & 1 & 0 \\ 0 & 0 & 1 & 4/c & -2/c & 1/c \end{array} \right], \text{ Hence inverse: } \frac{1}{c} \begin{bmatrix} -c-4 & c+2 & -1 \\ -2c & c & 0 \\ 4 & -2 & 1 \end{bmatrix}$$

$$R_0 = R_0 - R_2$$

Same steps but just treat as var, need $c \neq 0$ cond., also would show! poss.

Put in notes later!

$$z = B y \Rightarrow y = B^{-1} z$$

$$x = A B^{-1} z$$

Exercise 2.4.7 A

Given $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 & -1 & 2 \\ 1 & 0 & 4 \\ 2 & 1 & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$ and $\begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 1 \\ 2 & -3 & 0 \\ -1 & 1 & -2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$, express the variables

$x_1, x_2,$ and x_3 in terms of $z_1, z_2,$ and z_3 .

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 2 & -3 & 0 & 0 & 1 & 0 \\ -1 & 1 & -2 & 0 & 0 & 1 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & -1 & -2 & -2 & 1 & 0 \\ 0 & 0 & -1 & 1 & 0 & 1 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & 2 & -1 & 0 \\ 0 & 0 & -1 & 1 & 0 & 1 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|ccc} 1 & -1 & 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & 4 & -1 & 2 \\ 0 & 0 & -1 & -1 & 0 & -1 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 6 & -1 & 3 \\ 0 & 1 & 0 & 4 & -1 & 2 \\ 0 & 0 & 1 & -1 & 0 & -1 \end{array} \right]$$

$R_2 = R_2 - 2R_1$ $-R_2 = R_2$ $R_1 = R_1 - R_3$ $R_1 + R_2 \rightarrow R_1$
 $R_3 = R_1 + R_3$ $-R_3 = R_3$ $R_2 = R_2 - 2R_3$

$$\therefore B^{-1} = \begin{bmatrix} 6 & -1 & 3 \\ 4 & -1 & 2 \\ -1 & 0 & -1 \end{bmatrix}, \text{ Then want } AB^{-1}, \text{ so } \begin{bmatrix} 3 & -1 & 2 \\ 1 & 0 & 4 \\ 2 & 1 & 0 \end{bmatrix} \times \begin{bmatrix} 6 & -1 & 3 \\ 4 & -1 & 2 \\ -1 & 0 & -1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 18-4-2, -3+1+0, 9-2-2 \\ 6-4, 7, 3-4 \\ 12+4, -2-1, 6+2 \end{bmatrix} \Rightarrow \begin{bmatrix} 12 & -2 & 5 \\ 2 & -1 & -1 \\ 16 & -3 & 8 \end{bmatrix}, \text{ so } \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 12 & -2 & 5 \\ 2 & -1 & -1 \\ 16 & -3 & 8 \end{bmatrix} //$$